

SCHOOL MANAGEMENT APP

Project Requirements Specification

Version 1.0 | Confidential

Prepared for	School Management
Document type	Project Requirements Spec (PRS)
Platform	iOS · Android · Web Admin
Last updated	May 2026

1. Project Overview

This document defines the complete requirements for a School Management Mobile Application — a centralized platform connecting school management, teaching staff, administrative staff, students, parents, and prospective families under one secure, role-based system.

The app is designed to eliminate paper-based processes, improve parent engagement, streamline fee collection, and give school leadership real-time visibility into all operations. It will be built as a cross-platform mobile app (iOS and Android) with a separate web-based admin panel for desktop use.

1.1 Objectives

- Replace manual attendance registers, fee receipt books, and notice boards with a unified digital system
- Give parents real-time visibility into their child's attendance, homework, results, and fee status
- Enable teachers to manage their classroom digitally without relying on WhatsApp or phone calls
- Provide school management with live analytics and full operational control from a single dashboard
- Create a compelling guest experience that converts prospective parents into admissions enquiries

1.2 Scope

In scope	Out of scope
Mobile app — iOS and Android	Full LMS / e-learning platform (deferred)
Web admin panel for Super Admin and Admin Staff	AR campus tour (replace with video tour)
6 login roles with distinct dashboards	Student-to-student social posting
Online fee payment via Razorpay / PayU	Alumni network (year 3+)
Push notifications + SMS fallback	Inventory / stock management (optional add-on)
Live GPS bus tracking	Third-party ERP integration (future)
AI-powered features (v2)	

2. User Roles & Access Levels

The app has 6 distinct login roles. Each role has a completely separate dashboard and access level. The Super Admin assigns roles when creating user accounts.

2.1 Role Overview

Role	Who uses it	Access level
Super Admin	Principal / School Owner	Full control — all modules, settings, and users, leave approvals, daily dairy view (given by teacher who attended the class), school Expenses, Salary Categories, Salaries for the staff, Staff Management, Staff Attendance (view), Staff Access, full analysis of entire school and also app
Admin Staff	Office staff, accounts, coordinators	Student data, fees, attendance overview, daily dairy view (given by teacher who attended the class), designing the time table for the classes, assigning class teachers, design examination schedule and assigning staff(teachers) for exam invigilation duty, Staff Attendance (view), assign staff for substitute teacher if any teacher who's going to attend the class is on the leave
Teacher	Class and subject teachers	Own classes only — attendance, marks, homework, messaging, salary details, examinations, invigilation duty, Performance of students, daily dairy entry, leave management
Parent	Father / Mother / Guardian	Child's data — attendance, fees, results, bus tracking, fee payment, view fee structure of the child, time table of the class, holiday calendar, Exam schedule, parent's teacher meeting, daily dairy view
Guest	Prospective parents	Read-only — school info, faculty, facilities, fee structure

2.2 Super Admin — Full Control

Super Admin has exclusive access to:

- Create, edit, and delete all user accounts (staff, teachers, parents, students)
- Configure school branding: logo, name, color theme, contact details
- Set up academic years, terms / semesters, and holiday calendar
- Define and edit fee structure (tuition, transport, activity, exam fees per class)

- Enable or disable individual app modules school-wide
- View full analytics dashboard: attendance rates, fee collection, academic performance
- Send system-wide emergency broadcasts to all users simultaneously
- Export all data and manage cloud backup settings
- Approve or reject leave requests from all staff

2.3 Admin Staff — Operations & Student Data

Admin Staff is a sub-role within the Staff login. The Super Admin assigns this sub-role to front office staff and accounts team members. They have full access to student data but zero access to staff payroll, school finances, or system settings.

Admin Staff can access:

- Student enrollment and profile management (full student record)
- Admission number, class, section, date of admission, date of birth
- Student attendance percentage — monthly and overall
- Full fee structure for each student — term-wise breakdown
- Fee payment history — all transactions with receipt numbers
- Fee due / arrears — pending and overdue amounts with due dates
- Parent contact details — phone, address, alternate contact
- Student documents — transfer certificates, ID proof, certificates
- Send fee reminders via SMS or WhatsApp from the student profile
- Record cash fee payments on behalf of students
- Generate and print / share fee receipts as PDF
- Transport route management — bus routes, stops, driver details
- Library management — book catalogue, issue and return tracking
- Approve or reject leave applications from teaching and support staff
- Manage school calendar: events, exam dates, PTM schedules

Admin Staff CANNOT access:

- Staff salary, payroll, or HR records
- School-level financial reports or overall P&L
- Fee structure editing (view only — only Super Admin can edit)

- | |
|---|
| <ul style="list-style-type: none"> • Academic marks, gradebooks, or report cards |
| <ul style="list-style-type: none"> • User account creation or deletion |
| <ul style="list-style-type: none"> • App settings or system configuration |

2.4 Teacher Panel

- Mark daily class attendance: present / absent / late, with remarks
- Assign homework and assignments with due dates, file attachments, and auto-reminders
- Track homework submissions — see who submitted, who has not, and leave comments
- Enter exam and test marks — subject-wise, with grade auto-calculation
- Upload study materials: PDFs, videos, notes — accessible to students of their class
- Upload and manage lesson plans — internal reference, not visible to parents
- Post class-specific announcements and notices
- View personal teaching timetable — daily and weekly
- Schedule and manage class exams — date, subject, syllabus, duration
- Message parents of students in their class — direct, one-to-one
- Add virtual class links (Zoom / Google Meet) for online sessions
- Apply for leave — casual, sick, earned — with reason and document upload
- View leave balance and leave history

2.5 Parent Panel

- Multi-child support — one login manages multiple siblings with a child switcher
- Real-time attendance alerts: 'Vamshi entered school at 8:42 AM' (QR gate scan)
- View daily homework assigned to child
- View exam results, marks, and downloadable report cards
- Pay fees online — UPI, debit/credit card, net banking via Razorpay / PayU
- Download fee receipts as PDF
- Direct message to class teacher — in-app, no WhatsApp dependency
- View child's class timetable — daily and weekly
- Receive all school announcements, circulars, and notices
- Live GPS bus tracking — real-time map + ETA alert when bus is 5 minutes away
- Apply for student leave — date, reason, and supporting document upload
- View school gallery — events, competition results, student highlights (admin-posted)
- Submit complaints or feedback to school management
- View school calendar — holidays, exam dates, PTM dates, events
- View and respond to polls and surveys posted by the school (v3)

2.6 Guest Portal — Prospective Parents

The Guest portal requires no login. It is the school's 24/7 digital admissions brochure. Prospective parents download the app and immediately see:

- School overview — name, affiliation (CBSE/ICSE/State), ranking, founding year, pass rates, total students and faculty
- Infrastructure and facilities — labs, library, sports complex, transport fleet, canteen, medical room with photos
- Faculty showcase — name, designation, qualification, and years of experience for each teacher
- Achievements — awards, board exam toppers, competition results, news coverage
- Indicative fee structure — class-wise annual fees (clearly marked as indicative)
- Admissions information — process, eligibility, important dates, documents required
- Public notices — admission open announcements, upcoming events
- Enquiry form — parent name, mobile number, child name, class applying for, message
- School gallery — public photos and videos of events, campus, activities

Guest-to-Registration Conversion

- The guest portal ends with two clear calls-to-action: 'Submit Enquiry' (enquiry form sent to admin staff) and 'Create Parent Account' (begins the admission and registration process)
- A persistent banner throughout the guest experience shows: 'Register to access attendance, marks, and full parent features'
- All enquiries go directly into the Admin Staff dashboard as leads, with the parent's mobile number for follow-up

3. Core Modules

3.1 Attendance Management

Teacher — daily class attendance

- Teacher opens their class list and marks each student: Present / Absent / Late / Half-Day
- Once submitted, attendance is locked — editing requires Super Admin or Admin override
- Parent receives instant push notification: 'Vamshi was marked Absent on 29 May 2026'
- QR code attendance — student scans QR at school gate; system auto-marks present and notifies parent with entry time and photo
- Monthly attendance reports auto-generated — exportable as PDF or Excel
- Low attendance alert — admin staff automatically notified when a student's attendance drops below 75%

Staff attendance

- Admin staff and support staff clock in and out via the app
- GPS location captured at clock-in to prevent proxy attendance
- Monthly attendance summary auto-calculated for payroll reference
- Face recognition attendance at gate — hardware + software upgrade in v3

3.2 Homework & Assignment Module

- Teacher creates assignment: title, description, subject, class/section, due date, file attachments
- Students view assignments on their dashboard and submit digitally (PDF, image, or document)
- Auto-reminder push notification sent to student and parent 24 hours before due date
- Teacher dashboard shows: total assigned, submitted, pending, overdue — per assignment
- Teacher can leave comments and marks directly on each submission
- Parents see all homework assigned with submission status — submitted or pending
- AI homework explainer (v3): student clicks 'Explain this homework' and AI breaks it into simple steps

3.3 Exam, Results & Performance Analytics

- Admin staff creates exam: name, subject, date, duration, total marks, class/section
- Teacher enters marks after exam — subject-wise, student-wise entry
- System auto-calculates grade, percentage, class rank, and pass/fail status
- Report cards auto-generated and downloadable as PDF by parent and student

Performance analytics (not just marks)

- Subject-wise bar chart — strong and weak areas at a glance
- Rank trend graph — how the student's rank has moved over terms
- Comparison vs class average — above or below in each subject

- Weak subject flags — highlighted automatically based on marks pattern
- AI report card summary (v2): instead of 'Math: 72', shows 'Strong in geometry, needs more practice in algebra word problems'

3.4 Fee Management

- Super Admin defines fee structure: tuition, transport, activity, exam, library fees — per class, per term
- Admin staff views each student's fee status: structure, amount paid, amount due, overdue terms
- Online payment: parents pay via UPI, debit/credit card, or net banking through Razorpay or PayU
- Auto-generated PDF receipt sent to parent immediately after successful payment
- Cash payment recording: admin staff can log cash payments with receipt number
- Fee defaulter dashboard: list of students with outstanding dues, sorted by overdue amount
- Automated fee reminders: push notification + SMS sent 7 days before due date, on due date, and 3 days after
- Send targeted reminder from student profile: one-tap SMS or WhatsApp to parent
- Term-wise payment tracking: Term 1, Term 2, Term 3 — paid / pending / overdue status
- Export fee collection reports: daily, monthly, class-wise — PDF and Excel
- Super admin have authority to give Fee Concession for the student fee.

3.5 Communication

In-app messaging

- Direct one-to-one chat: Teacher to Parent (of students in their class only)
- Admin staff to Parent: for fee-related and administrative queries
- Super Admin can message any user in the system
- Message history retained — no conversation lost if app is reinstalled

Announcements & circulars

- Super Admin and Admin Staff post circulars: school-wide or class-specific
- Teacher posts class notices: visible only to parents of their class
- All announcements appear in a chronological feed with date and sender
- Document attachments supported: PDF circulars, forms, permission slips

Emergency alert system

- One-tap broadcast by Super Admin to ALL parents and staff simultaneously
- Delivery: push notification (primary) + SMS fallback (for parents with notifications off)
- Alert types: sudden holiday, early dismissal, weather alert, school emergency, exam postponement
- Alert history retained — parents can re-read past alerts in the app

Push notifications — all roles

- Attendance marked / entry time alert (parent)
- New homework assigned (student and parent)
- Homework due date reminder — 24 hours before (student and parent)
- Fee due reminder — 7 days before, on due date, 3 days after (parent)
- Fee payment confirmed (parent)
- New marks entered / report card available (parent and student)
- New announcement or circular posted (relevant role)
- Leave request status updated: approved or rejected with remark (staff)
- New leave request submitted (admin / super admin)
- Bus ETA alert: 5 minutes away from child's stop (parent)

3.6 Timetable & School Calendar

- Super Admin or Admin Staff creates and publishes class-wise timetables
- Teacher views personal teaching timetable — daily and weekly view
- Student and parent view child's class timetable — daily and weekly view
- School calendar: all users see upcoming events, exam dates, holiday list, PTM dates
- Exam schedule: date, subject, timing, venue — auto-shared to relevant classes
- Parent-Teacher Meeting scheduler : admin/admin staff confirms
- Live updates: timetable or exam schedule changes reflected immediately with push notification

3.7 Leave Management

Staff leave request

- Staff selects leave type: Casual Leave (CL), Sick Leave (SL), Earned / Privilege Leave (EL)
- Selects from and to dates — system auto-calculates working days, excludes Sundays and holidays
- Enters reason and optionally uploads a document (medical certificate etc.)
- Live leave balance shown before submission: CL / SL / EL remaining days
- Submitted request shows in staff's leave history as 'Pending'
- Staff can cancel a pending request before it is actioned

Admin / Super Admin approval

- Pending requests appear in admin dashboard with push notification badge
- Admin views: staff name, designation, leave type, dates, reason, and attached document
- Admin can add a remark (mandatory if rejecting — staff see this reason)
- One-tap Approve or Reject with confirmation prompt
- On approval: leave deducted from balance, added to school calendar, staff notified

- On rejection: staff notified with admin remark, balance unchanged
- Team calendar view: shows all approved leaves — admin can spot coverage gaps before approving
- Export: monthly leave report per staff member — PDF and Excel

3.8 Digital ID Card

- Auto-generated for every student on admission
- Contains: student photo, full name, admission number, class and section, blood group, emergency contact number, bus route number, academic year
- QR code links to student's school profile — scannable by gate staff and admin
- Downloadable as PDF — parent can print or show on phone at gate
- Staff digital ID: name, designation, employee ID, department, emergency contact, QR code

3.9 Transport & GPS Bus Tracking

- Admin staff enters bus routes: route name, stops in order, driver name and mobile number, vehicle number, capacity
- GPS device installed in each bus transmits real-time location every 10–15 seconds
- Parent sees live bus location on map within the app
- ETA calculated dynamically — updates as bus moves
- Push alert sent to parent when bus is 5 minutes from child's stop (morning pickup and evening drop)
- Driver can send a delay alert from their phone: 'Bus delayed by 20 minutes due to traffic'
- Admin staff can view all buses on a fleet map — real-time overview

3.10 Library Management (Version 2)

- Admin staff or librarian maintains book catalogue: title, author, ISBN, quantity, category
- Issue a book to a student: admission number lookup, book selected, due date set
- Student and parent see issued books and due dates in their app
- Auto-reminder 3 days before book return due date
- Mark book returned — inventory auto-updated
- Overdue list: all students with unreturned books past due date

3.11 School Social Feed (Version 2)

- Admin-controlled only — no student or parent posting (prevents moderation issues)
- Admin posts: school events, competition results, student achievements, photos, videos
- All parents and students see the feed — school news in one place
- Student spotlight: admin highlights individual student achievements with photo and caption
- Gallery: organised albums by event — sports day, annual day, science fair, field trips